

Tide retrospective: mixed quartic-sextic 2D partition (A2)

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Abstract

A trivial-mirror tide of A3 (separable-potential abstraction, Tide 12). For the 2D pure-monomial separable potential $L(x, y) = x^4/24 + y^6/720$, the partition function and moments factor as products of the corresponding 1D quartic and sextic quantities. Closed-form value of the 2D partition function:

$$Z_{2D}(t) = \frac{1}{2}(24/t)^{1/4}\Gamma(1/4) \cdot \frac{1}{3}(720/t)^{1/6}\Gamma(1/6).$$

Single session, 153 lines in one new file (`Laplace/TwoD/QuarticSextic.lean`), zero `sorry`, zero `axiom`, zero `native_decide`. Five public theorems (partition factorisation, closed form, partition positivity, moment factorisation, mixed-coordinate covariance vanishing); all are `apply-and-discharge` chains over the existing A3 theorems plus integrability witnesses from the existing pure-monomial infrastructure. No GPT consult was needed: there is no candidate-design surface, since the target is the A3 theorem family instantiated.

1 Setting

The A-category in the cross-project candidates survey carried three follow-ups after Tide 12 (the A3 separable-potential abstraction) landed: A1 (generic even-monomial 1D template), A2 (quartic-sextic mixed 2D), and A4 (generic mixed even-power 2D moments). A2 is the smallest and most concrete: a single 2D potential with two explicitly-named factors, both of which already have closed-form partition functions in the laplace repo (`Laplace.OneD.quartic_partition` and `Laplace.OneD.sextic_partition`). The candidates survey estimated ~ 30 lines.

The seabed at the start of the tide was `main` at commit `b844d7a`¹, with the I2 algebra (Tide `tide/gibbscov-algebra`) just merged in. All ingredients for A2 were on `main`: the pure-monomial potentials and their positivity / integrability lemmas in `OneD/Quartic.lean` and `OneD/Sextic.lean`, and the generic A3 theorem family in `TwoD/AddSeparable.lean`.

2 The thing formalised

Five theorems in the new file `Laplace/TwoD/QuarticSextic.lean`, all in the `Laplace.TwoD` namespace:

- `partitionFunction_quarticSextic_factor`: $Z_{2D}(t) = Z_{\text{quartic}}(t) \cdot Z_{\text{sextic}}(t)$ for $t > 0$.

¹Commit prefixes truncated to seven characters throughout this retrospective; sufficient for uniqueness across the laplace history.

- `partitionFunction_quarticSextic_eq`: the explicit closed-form value, obtained by substituting `quartic_partition` and `sextic_partition`.
- `partitionFunction_quarticSextic_pos`: strict positivity, by `mul_pos` on the two pure-monomial positivity lemmas.
- `gibbsExpectation_quarticSextic_factor`: for any separable observable $f(x) \cdot g(y)$ with appropriate integrability, $\langle f(x)g(y) \rangle_{2D,t} = \langle f \rangle_{\text{quartic},t} \cdot \langle g \rangle_{\text{sextic},t}$.
- `gibbsCov_quarticSexticfst_snd_eq_zero`: the covariance of any first-coordinate observable $f(x)$ and any second-coordinate observable $g(y)$ against the 2D quartic-sextic Gibbs measure is 0. The structural statement of independence under the separable measure.

The informal content is exactly what the A3 abstraction says, applied to one specific pair of potentials. The added value is twofold: (a) the closed-form value of the 2D partition is now a one-rewrite substitution rather than a chain through generic infrastructure, and (b) the moment-factorisation and covariance-vanishing theorems take *integrability hypotheses against the 1D potentials* directly, which is what every downstream consumer would need to phrase anyway.

3 Proof strategy

Each public theorem is a single `apply` of the corresponding A3 generic theorem, with the integrability hypotheses discharged from the existing pure-monomial infrastructure (`quartic_integrable_pow_pot` at $n = 0$ for the quartic factor; `sextic_integrable_pow_pot` at $n = 0$ for the sextic factor). The two integrability discharges are factored into private helpers (`integrable_exp_neg_t_quartic`, `integrable_exp_neg_t_sextic`) so the public proofs are clean.

For the closed-form theorem, after the factorisation rewrite the proof is a single `rw` chain combining the factorisation lemma with the two pure-monomial closed forms² – nothing else needed, because the pure-monomial closed-form expressions are already exactly the form we want.

4 Roadblocks and resolutions

None. The proof rolled through in a single pass with no re-architecting, no thrashing, no Mathlib-search round. The most substantive design decision was whether to make the integrability helpers private to this file or promote them to the `AddSeparable` infrastructure file: kept them private here, since they’re tied to the specific (quartic, sextic) instantiation and the `integrable_pow_pot 0 ht` pattern is generic enough that any future similar tide can reproduce it in two lines.

5 What was Mathlib, what was new

Mathlib lookups. None new beyond `ne_of_gt` and `mul_pos`.

laplace-side existing material reused.

²`partitionFunction_quarticSextic_eq`; the chain is `rw [partitionFunction_quarticSextic_factor ht, quartic_partition ht, sextic_partition ht]`.

- The three A3 generic theorems from `TwoD/AddSeparable.lean` (Tide 12): the partition factorisation, the moment factorisation for separable observables, and the mixed-coordinate covariance vanishing.³
- Pure-monomial closed forms and their positivity / integrability lemmas in `OneD/Quartic.lean` and `OneD/Sextic.lean`.⁴

New material. Two private integrability helpers and five public theorems (partition factorisation, partition closed form, partition positivity, moment factorisation, mixed-coordinate covariance vanishing). Total: 153 lines, of which ~ 60 are docstrings and ~ 90 are proof body.

6 Lessons

Trivial-mirror tides are valuable. A2 is the second trivial-mirror tide of the day (after L1 sextic-J-function on 6 May; today’s similar mirrors include G2/G2+ kappa3-affine-anharmonic). The pattern is: a generic theorem family lands; a strict-improvement tide instantiates it at specific names. The instantiation is a 50–150 line file, no new mathematics, all proof-engineering value. These tides are *worth doing* because they (a) surface the closed-form values that downstream consumers will want, (b) front-load the integrability bookkeeping at the named-instance level rather than forcing every consumer to repeat it, and (c) provide a stable named target for paper-side `\leanref` tags.

No-deliberation tides skip GPT cleanly. The skill’s “trivial mirror” exemption was used here for the second time today (after L1 sextic-J-function and G2 anharmonic-affine mirrors). The criterion is honest: there must be *no candidate choice* – the target is forced by the abstraction, the proof is a single `apply`-chain, and the integrability hypotheses are already established. Deliberation overhead would not have surfaced anything useful. Worth being explicit: the exemption is for tides where the deliberation surface is genuinely zero, not for tides where we’re *lazy* about deliberation.

7 Follow-ups

- *A4 – mixed even-power 2D moments.* The natural sequel: the closed form of $\langle x^{2j} y^{2k} \rangle_{2D,t}$ under the same potential, by applying `gibbsExpectation_quarticSextic_factor` to $f = x^{2j}$, $g = y^{2k}$, then substituting `quartic_expected_value_even` and `sextic_expected_value_even`. Estimated ~ 30 lines.
- *A1 – generic even-monomial 1D template $L = x^{2k}/(2k)!$.* The natural *abstraction* step: unify the quartic, sextic, and (now) generic- k J-function machinery under a single parametric family. Estimated ~ 250 – 350 lines per the candidates survey.
- *Other 2D mixed pure-monomial pairs.* Quartic-quartic, sextic-sextic, $x^{2j}/((2j)!)$ paired with $y^{2k}/((2k)!)$ for various (j, k) . Each is a 30-line corollary of A3 once the named 1D pure-monomial families exist. Best deferred until A1 lands and gives the parametric 1D pattern.

³Names: `partitionFunction_addSeparable_factor`, `gibbsExpectation_separable_addSeparable`, `gibbsCov_addSeparablefst_snd_eq_zero`.

⁴Six lemmas: `quartic_partition`, `quartic_partition_pos`, `quartic_integrable_pow_pot`, `sextic_partition`, `sextic_partition_pos`, `sextic_integrable_pow_pot`.

- `\leanref tag for the primer`. The 2D mixed quartic-sextic example may appear in the susceptibility primer’s exposition; if so, tag with `\laplaceCommit` pinned to the merge commit. Process tide.

Acknowledgements

This Tide consumed only material on `main` at the time of branching (commit `b844d7a`, post-I2-merge). It is genuinely non-rivalrous with the half-dozen other tides running concurrently: A2 only writes a new file, doesn’t touch any existing module, and its only dependency on today’s earlier work is the I2 algebra (transitively, via `Laplace.gibbsExpectation.const` inside the A3 theorems’ proofs).

GPT-5.5 Pro was not consulted: the trivial-mirror exemption applies, since there are no candidate-design choices and the proof is a direct **apply**-chain over the A3 theorem family. This is the second use of that exemption today (the first was the L1 sextic-J-function mirror tide on 6 May).